reflectionschema in set-theory/replacement

The Open Logic Project contributors

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Editorial note. All mathematical statement and proof text is reproduced verbatim from the Open Logic Project, unit set-theory, file finiteaxiomatizability.tex, fixed at commit 1e960beff9ed7835bf3e3f1335e21af3439cd107. Only the packaging (the adapter preamble and the standalone wrapper) is this module's.

Editorial note. Numbering inside this standalone excerpt is editorial, not the Open Logic Project print numbering.

Editorial note. The adapter preamble loads the project's OWN style layer verbatim (sty/open-logic-formulas.sty, sty/open-logic-tokenize.sty, sty/open-logic-selective.sty and open-logic-config.sty, including its tag truth assignment), so the author's text-token shorthand, !A math metavariables, rule names (Intro, Elim, FalseInt) and the tag conditionals resolve exactly as in the book. Three substitutions are forced by the pinned toolchain and are recorded: isomath.sty is absent so \mathsfit is \mathsf; xpunctuate.sty is absent and unused so its RequirePackage is dropped; thmtools/ntheorem are absent so the environments are amsthm with the project's family numbering. The proof trees are the project's own bussproofs-extra.

Editorial note. The H2 judgement and its reason were made in this module by reading the author's own statement and complete proof at the pinned line span. Original candidate key: set-theory/replacement|finiteaxiomatizability.tex|zfnotfinitely.

Editorial note. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites. The author's own ollabel is kept unprefixed (not part:chapter:section) so those pointers resolve; the label text itself is the author's own argument.

Editorial note. The author's own text names 1 pointer(s) that this module could not resolve to a definition site in the pinned checkout and that are therefore not registered: sth:replacement:ref:reflectionschema.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own olref pointers. Its prerequisites are the earlier results of the same pinned Open Logic Project commit that the author cites there.

Theorem 0.1. ZF *is not finitely axiomatizable. More generally: if* $\mathbf{T}$ *is finite and* $\mathbf{T} \vdash \mathbf{ZF}$*, then* $\mathbf{T}$ *is inconsistent.*

(Here, we tacitly restrict ourselves to first-order sentences whose only non-logical primitive is $\in$*, and we write* $\mathbf{T} \vdash \mathbf{ZF}$ *to indicate that* $\mathbf{T} \vdash \varphi$ *for all* $\varphi \in \mathbf{ZF}$*.)*

Proof. Fix finite $\mathbf{T}$ such that $\mathbf{T} \vdash \mathbf{ZF}$. So, $\mathbf{T}$ proves Reflection, i.e. ??. Since $\mathbf{T}$ is finite, we can rewrite it as a single conjunction, θ. Reflecting with this formula, $\mathbf{T} \vdash \exists\beta(\theta \leftrightarrow \theta^{V_\beta})$. Since trivially $\mathbf{T} \vdash \theta$, we find that $\mathbf{T} \vdash \exists\beta\ \theta^{V_\beta}$.

Now, let $\psi(X)$ abbreviate:

$$\theta^X \wedge X \text{ is transitive} \wedge (\forall Y \in X)(Y \text{ is transitive} \rightarrow \neg\theta^Y)$$

roughly this says: X is a transitive model of θ, and $\in$-minimal in this regard. Now, recalling that $\mathbf{T} \vdash \exists\beta\ \theta^{V_\beta}$, by basic facts about ranks within **ZF** and hence within $\mathbf{T}$, we have:

$$\mathbf{T} \vdash \exists M\psi(M). \tag{*}$$

Using the first conjunct of $\psi(X)$, whenever $\mathbf{T} \vdash \sigma$, we have that $\mathbf{T} \vdash \forall X(\psi(X) \rightarrow \sigma^X)$. So, by (*):

$$\mathbf{T} \vdash \forall X(\psi(X) \rightarrow (\exists N\psi(N))^X)$$

Using this, and (*) again:

$$\mathbf{T} \vdash \exists M(\psi(M) \wedge (\exists N\psi(N))^M)$$

In particular, then:

$$\mathbf{T} \vdash \exists M(\psi(M) \wedge (\exists N \in M)((N \text{ is transitive})^N \wedge (\theta^N)^M))$$

So, by elementary reasoning concerning transitivity:

$$\mathbf{T} \vdash \exists M(\psi(M) \wedge (\exists N \in M)(N \text{ is transitive} \wedge \theta^N))$$

So that $\mathbf{T}$ is inconsistent.[1] □

[1] This "elementary reasoning" involves proving certain "absoluteness facts" for transitive sets.